Answer Key - Computer Science - K-12 Level Assessment 15 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) 3; B) 4; C) 2; D) 8
   Note: The expression x >> 1 in Python 3 performs a bitwise right shift on the binary representation of x, effectively dividing the number by 2, so shifting the binary representation of 8 (which is 1000 in binary) to the right by one position results in 4 (which is 0100 in binary).

2. **Answer:** B
   Options: A) A device driver is assigned to the device.; B) An Internet Protocol (IP) address is assigned to the device.; C) A packet number is assigned to the device.; D) A Web site is assigned to the device.
   Note: When a new device is connected to the Internet, it is assigned an Internet Protocol (IP) address, which uniquely identifies it on the network and allows for communication with other devices.

3. **Answer:** D
   Options: A) Error; B) a; C) b; D) c
   Note: In Python, negative indexing allows access to elements from the end of a string, so "abc"[-1] retrieves the last character, which is 'c'.

4. **Answer:** B
   Options: A) replace(old, new [, max]); B) strip([chars]); C) swapcase(); D) title()
   Note: The 'strip()' function in Python 3 is specifically designed to remove all leading and trailing whitespace characters from a string, making it the correct choice for this question.

5. **Answer:** C
   Options: A) The goal of the attack; B) The number of computers being attacked; C) The number of computers launching the attack; D) The time period in which the attack occurs
   Note: A distributed denial-of-service (DDoS) attack involves multiple compromised computers working together to overwhelm a target, whereas a denial-of-service (DoS) attack typically originates from a single source, highlighting the key difference in the number of attacking computers.

True / False

6. **Answer:** False
   Options: A) True; B) False

Note: The statement is false because the expression ['Hi!'] * 4 is valid syntax in Python 3 and results in a list containing four copies of the string "Hi!".

7. **Answer:** True
Options: A) True; B) False
Note: O(1) indicates that the time taken by an algorithm does not depend on the size of the input, making it the fastest possible time complexity and thus the smallest in the hierarchy of asymptotic notations.

8. **Answer:** False
Options: A) True; B) False
Note: The statement is false because O(log n) represents logarithmic growth, which increases much more slowly than O(n^2), indicating that as n increases, O(n^2) grows significantly faster than O(log n).

9. **Answer:** False
Options: A) True; B) False
Note: In Python 3, floor division is performed using the operator '//' instead of the operator '/', which is used for regular division.

10. **Answer:** True
Options: A) True; B) False
Note: Sorting values in a column or row organizes the data in a way that makes it easier to spot outliers or anomalies, such as unusually high or low entries, compared to the rest of the dataset.

Long Form Response

11. **Answer:** When we process the list l = [1, 2, 2, 3, 4] using the set data structure in Python 3, the output is {1, 2, 3, 4}. A set automatically removes any duplicate values, which is why the repeated number '2' appears only once in the result. This feature is significant because it allows us to obtain a collection of unique elements from the list, making it easier to analyze data without redundancy. The use of sets is particularly useful in various applications, such as counting distinct items or performing operations like unions and intersections efficiently.
Note: correct because it accurately describes how the set data structure in Python 3 removes duplicate values from the list, resulting in a collection of unique elements, which is essential for data analysis and various computational applications.

12. **Answer:** The trend of computer processing speeds doubling approximately every two years significantly influences technology companies' research and development strategies. Companies can set ambitious goals for their projects based on the expectation that faster processing will allow for more complex and innovative applications. For example, a company focusing on artificial intelligence might plan to develop more sophisticated algorithms that can leverage the increased speed for quicker data analysis.

This foresight encourages investment in cutting-edge technologies and can lead to advancements in various fields, such as gaming or data science, where processing power is crucial. By anticipating these changes, companies can stay competitive and push the boundaries of what is possible with technology.
Note: The answer correctly highlights how the predictable increase in processing speeds enables technology companies to set ambitious research and development goals, allowing them to innovate and create more complex applications, such as advanced artificial intelligence algorithms that benefit from faster data processing.

End of Answer Key